

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science and Engineering (CSE).

Semester – 3rd. Section- 1, 2 ,3 & 4

Mid-Term Examination September– 2026

Subject Name/ Code:- Signal and Communication Systems CSE/CSE(AI)/CSE(DS)/ CSE(CPS)/ CSE(CS)-3005

Max. Marks: 20

Time Allotted: 90 min

NOTE: 1. All questions are compulsory.

2. Mark answers number clearly.

3. Draw neat and clean sketches wherever needed

S. No	Question	Marks
1.	(a) Consider the system, $y(t) = x^2(t) + x(t-2)$. Investigate the system for linearity/Non-Linearity, Causal/Non-Causal, and time variant / invariant using the mathematical definitions of the properties.	3
	(b) Consider the signal, $x(t) = \cos(6t) + \cos(10t) + \sin(14t).$ Establish whether the signal is periodic. If periodic, obtain its fundamental angular frequency ω_0 and fundamental period T_0 . Also express each frequency component in terms of the fundamental harmonic.	2
2.	(a) Comment on Dirichlet's conditions for existence of Fourier Series.	1
	(b) Consider a ramp Signal, $r(t) = t$, for the interval of 0 to 2π . Find the Trigonometric Fourier series of $r(t)$.	2
	(c) An LTI system has impulse response, $h(t) = e^{-2t} \cdot u(t)$ and the input is $x(t) = u(t) - u(t-1)$. Obtain the response of a system, $y(t)$. Support your answer with Sketches	2
3.	(a) Find the Fourier transform of following signals: (i) Rectangular Pulse (ii) Unit Step Function	2
	(b) If, $X(\omega) = \delta(\omega - \omega_0)$, then $x(t)$ is ?	1
	(c) Given the Fourier Transform pair, $x(t) \leftrightarrow X(\omega)$, derive the Fourier Transform of $y(t) = \frac{d}{dt} [x(2t - 4)].$ Obtain $Y(\omega)$, completely in terms of $X(\omega)$, clearly applying the appropriate transform properties.	2
4.	(a) State and Prove Time Shifting Property and Frequency Differentiation Property of Laplace Transform	3
	(b) Obtain $X(s)$, the Laplace Transform, of a sine wave which is given as, $x(t) = A \sin \omega_0 t \cdot u(t).$ And specify the ROC.	2
